Harish G

Agentic AI & Automation Engineer | LLM Guardrails, Tool Calling, RAG, Agent Orchestration

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Profile

I do not trust model output, and I build like it. Every tool an agent can reach is a declared function, every value it extracts is coerced to its declared type before anything downstream reads it, and a number the model states that no tool produced gets marked unverified rather than shown. I am a Software Developer at CortexCraft.ai, where that discipline runs a fleet of Gemini Live phone agents in production.

Core Skills

Languages	Python · JavaScript · TypeScript · Java · Kotlin · SQL
AI / ML	LLM Integration · RAG · Agents & Tool Calling · Output Guardrails · Context Management · Gemini Live · STT/TTS Pipelines · Prompt Frameworks
Backend	FastAPI · Node.js/Express · Spring Boot · Celery · Alembic · WebSockets · REST API Design
Data	PostgreSQL · MongoDB · Neo4j · Supabase/pgvector · Redis · Firebase
Platform	Twilio · Plivo · WhatsApp Business API · Docker & Compose · Jenkins · GitLab CI
Concepts	System Design · Modular Architecture · JWT Auth & RBAC · Multi-Tenant SaaS · Query Optimization

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Built and maintain the voice-agent platform (589 of 626 commits on the SaaS platform, 67 of 75 on the core engine): Gemini Live phone agents over Twilio and Plivo, with per-agent RAG grounding, prompt generation, transcripts and an external REST API that other products call instead of forking the stack.
- Design the guardrails and tool-calling orchestration that constrain model output. Tools are declared functions. Extracted call data is coerced to its declared type before anything downstream trusts it, and escalation to a human is an explicit path rather than a fallback.
- Shipped LeadCall AI on that platform: multi-tenant outbound calling with Celery dial and callback workers, pitch-document RAG and Alembic-migrated PostgreSQL behind a React dashboard.
- Own production reliability of the call pipeline. Fixed outbound routing, added reconnect handling for dropped Gemini sockets, and traced a Docker TLS failure to NAT hairpinning.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

- Delivered full-stack web and app projects end to end for local clients: React/Vite front ends, Python and Node.js services and automation scripts, plus ongoing fixes and deployment support.

Selected Projects

AI Personal CFO and Talo *Kotlin, Jetpack Compose, Hilt, Room/SQLCipher, ML Kit, Gemini Live*

- Sole author, 159 commits, 34 Gradle modules, 199 test files. Twelve deterministic engines compute every figure and the model is only permitted to explain them, on a written rule: numbers from math, words from AI.
- An output guardrail rewrites any number in model text that is absent from the tool result as `[unverified]`. An MCP-style in-process tool host exposes the engines through a JSON registry carrying `inputSchema`, `readOnlyHint` and `destructiveHint`; chat reads through the registry and never writes.

Dental Clinic AI Assistant *Python, Flask, PostgreSQL/pgvector, Gemini, WhatsApp Business API*

- Chunked document RAG (800-token chunks, 100 overlap) embedded with `text-embedding-004` into pgvector, retrieved by cosine distance with a full-text fallback and role-scoped row filtering applied inside the retriever, so authorisation cannot be prompted around.
- Gemini function calling for booking behind a role-based tool router admitting one tool per turn, and a 19-state WhatsApp conversation state machine that hands off to a phone agent with context preserved.

KisanVoice AI *Python, FastAPI, React, MongoDB, WhatsApp Flows, Google STT/TTS, Jenkins*

- 172 of 192 commits. Voice and text surveys in 24 configured locales (10 Indian) from a config-driven language registry, so adding one is a config row rather than a deploy; option matching survives imperfect speech through fuzzy and semantic fallback.
- Migrated the platform off its Node backend onto FastAPI and off Sarvam onto Google STT/TTS without interrupting live surveys.

JiraFlow Agent *Python, React, Jira API, GitHub API, LLM Agents*

- Turns a Jira ticket into a branch, a code change and a pull request, one `ai/scrum-<id>` branch per ticket. The agent proposes and a human merges, so a wrong edit costs a rejected PR rather than a broken main branch.

Education & Ways of Working

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore

Feature branches and conventional commits; GitLab CI with 132 pytest files as the merge gate on the voice-agent platform.